

PARAMETER-EFFICIENT VISION-LANGUAGE MODEL FINE-TUNING FOR MULTITASK BIOAVAILABLE IRON PREDICTION

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ABSTRACT

[Abstract...]

1. INTRODUCTION AND RESEARCH BACKGROUND

Iron deficiency, either iron deficiency anemia or iron deficiency without anemia, has been estimated to affect over 3 billion people worldwide and cause debilitating symptoms like fatigue and exercise intolerance (Al-Naseem et al., 2021). After being diagnosed with an iron deficiency, I had to actively increase my iron not only through supplements, but through diet as well. This drove me to learn more about the iron content of different foods, factors affecting iron absorption, and challenges of increasing iron through diet.

The total iron present in food differs from the amount of iron the body can absorb and use, which is called bioavailable iron (Hallberg & Hulthén, 2000). In other words, how much iron the body absorbs is less than the iron measurement provided in nutrition labels and nutritional data sources like USDA FoodData Central. Additionally, there are two distinct categories of iron: heme iron, which comes from meat, and non-heme iron, which comes from plants. Heme iron is better absorbed than non-heme iron and different ingredients can influence the absorption of non-heme iron within a dish. For instance, calcium sources like cheese can inhibit iron absorption, whereas vitamin C sources like bell peppers can enhance it (Hallberg & Hulthén, 2000).

Most food image analysis models and apps provide estimated measurements of calories and macronutrients like fat, protein, and carbohydrates, but not micronutrients like iron. Those that do provide iron estimates have measurements based on total iron, not bioavailable iron. Bioavailable iron provides a better estimate because it considers how much of the iron is heme and/or non-heme as well as whether iron absorption inhibitors, like cheese or cream, or enhancers, like peppers or broccoli, are also present in the food. I intend to fill the gap by fine-tuning a VLM (Vision Language Model) to analyze food images and provide estimates of bioavailable iron and ingredient portion sizes for reproducing bioavailable iron calculations. Previous work has been done to fine-tune a VLM for calorie estimation (Yao et al., 2024), but not for bioavailable iron or ingredient portion sizes.

2. RESEARCH AND METHODS

First, I will need to do some data engineering and query the USDA FoodData Central API to get heme/non-heme iron (which is to be determined based on whether the ingredient is meat or not), vitamin C, and calcium measurements for each of the ground truth ingredient portions from my subset of food images. I will use this information to calculate the ground truth bioavailable iron per image.

Next, to create the training, validation, and test data, I will generate question-answer pairs like the example shown below:

```
[
  {
    "question": "How many mg of bioavailable iron is in this food?",
    "answer": "2.6 mg",
    "image_file": "train/7843322356500104680_443548_.jpeg"
  },
  {
    "question": "Can you list the ingredient portions in this food in decreasing order?",
```

```
"answer": "300g chicken, 50g egg"

"image_file": "train/7843322356500104680_443548_.jpeg"

},

...

]
```

Then, I will fine-tune a pretrained Vision Language Model, SmolVLM, using PEFT and LoRA techniques with GPU(s) on Google Colab Pro.

Lastly, I will evaluate my model with the metrics I discuss in the next section.

3. MATERIALS AND DATA SOURCES

I plan to use a stratified subset (at least 5K images) of the [MM-Food-100K](#) dataset, which includes a wide diversity of homemade, restaurant, and packaged food and produce, as well as ground truth ingredient portions like: ["bread:100g", "cheese:30g", "egg:50g", "sausage:50g"] (Dong et al., 2025). Because the dataset includes measurements of only macronutrients and not micronutrients, I will also query the [USDA FoodData Central API](#) to get total iron, vitamin C, and calcium per ground truth ingredient in my subset of the MM-Food-100K dataset.

4. RESULTS

The results were...

5. DISCUSSION AND CONCLUSION

Discuss the conclusion...

REFERENCES

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